

To: The Editorial College
SciPost Physics
Stichting SciPost, Amsterdam, The Netherlands
September 18, 2026

Subject: Submission of Original Research Manuscript

Title: *Mechanized T-Duality and Frontier String Dynamics on $K3 \times T^2$: Closing the Loop via LeanFlow Stiff Solvers with Native Duality, Topological Data Analysis, and Lean 4 Model Verification*

Dear Editors of SciPost Physics,

We submit our original research manuscript for consideration as a Regular Research Article in *SciPost Physics*.

A longstanding challenge in string theory and non-linear cosmology is that numerical solvers inevitably crash near self-dual radii ($R \approx \sqrt{\alpha'}$) or singular orbifold points due to artificial coordinate divergences. Furthermore, verifying that dynamical defect networks formed during cosmic phase transitions satisfy global quantum consistency constraints—such as Ramond-Ramond tadpole cancellation ($\sum Q_i = 0$)—has historically relied on manual, error-prone bookkeeping.

In this work, we present **LeanFlow**: a duality-aware neuro-symbolic framework coupling high-performance stiff numerical integration with discrete algebraic invariant checks in the Lean 4 interactive theorem prover. Following this submission’s internal audit, we label every claim below by evidentiary tier (A: kernel-checked and adequate; B: exact arithmetic with a negative control; C: conjecture/interpretation; X: exploratory numerics), and we have revised several claims from the originally deposited version accordingly. Our manuscript couples numerical integration with post-hoc topological analysis (Numerical Integration $\rightarrow$ Topological Extraction via TDA Mapper $\rightarrow$ Lean discrete-consistency checking); Lean verifies discrete consistency checks algebraically, though these checks rest on hard-coded model assumptions (tier C), and delivers the following primary physical and methodological advances:

- 1. Native Duality Engine & Algorithmic Section Condition:** LeanFlow embeds continuous Buscher symmetry locks ($R \leftrightarrow \alpha'/R$) and modular group $SL(2, \mathbb{Z})$ invariance directly into an adaptive Backward Differentiation Formula (BDF) solver. By co-simulating dual geometric frames (Type IIA $\leftrightarrow$ Type IIB), the solver circumvents coordinate singularities as the cosmological modulus relaxes from high energies toward the Fricke saddle ($\tau = i$) and Kummer orbifold fixed point ($\tau = e^{i\pi/3}$) without metric breakdown (tier X; the numerical minimum reported in an earlier draft could not be traced to any committed simulation output and has been removed pending a reproducible source).
- 2. Closed-Loop Defect Verification:** Passing the stochastic Langevin dynamics of a multi-well landscape across thermal symmetry breaking ($T_c = 1.0$) into a Topological Data Analysis (TDA) Mapper 1-skeleton graph, we partition the network into string-theoretic equivalence classes (Attractor Vacua, Domain Walls, and Cosmic Strings). Node/edge counts and the graph’s cycle rank β_1 vary between runs because the Mapper stage is not seeded, so we report this construction as tier X exploratory numerics rather than quote a single fixed count; β_1 here denotes the cycle rank of the Mapper graph, not a persistent-homology invariant. As a discrete algebraic consistency check (tier C), Lean 4 verifies that, given the hard-coded per-class charge assignments, the mapped equivalence classes satisfy Ramond-Ramond tadpole cancellation ($\sum Q_i = 0$); this checks internal consistency of the labeling scheme, not an independent physical derivation.

3. **Frontier Non-Perturbative Dynamics:** We apply the pipeline to three core open problems in string theory:

- *Swampland Distance Conjecture:* Moduli geodesic flows on $K3 \times T^2$ are computed, with H_0 barcodes tracking the collapse of the Kaluza-Klein mass tower ($M \leq M_0 e^{-\alpha \Delta d}$), where $\alpha = 1/\sqrt{2}$ is an imposed, hard-coded model input rather than a fitted or measured quantity, and the committed run’s bound check carried $1.5\times$ slack rather than a strict tolerance (tier X).
- *Sen’s Tachyon Condensation:* Non-linear brane-antibrane roll-down is simulated, extracting the localized BPS soliton via TDA Mapper. As a discrete algebraic consistency check (tier C, not an independent derivation), Lean 4 checks Grothendieck group charge conservation $[E] - [F] = [D_{\text{defect}}]$ given the modeled bundle classes assigned.
- *Vacuum Decay in the Flux Landscape:* We explore vacuum decay using flat-space bounce dynamics solved via shooting methods; gravity is omitted, so this is not a Coleman–De Luccia bounce in the strict sense. The central charge $c(N) = 100(N+1)$ is a hand-chosen exploratory toy model (tier C, we conjecture it as a qualitative proxy), and Lean 4 checks monotonicity of this hand-picked formula rather than deriving a holographic c -theorem. Full Coleman–De Luccia dynamics with gravity is left to future work.

4. **Cosmological Stability & Energy Conditions:** We solve the non-linear Poisson boundary value problem for Symmetron screening, obtaining a screening factor $\Delta R/R \approx 1.18 \times 10^{-4}$ from the committed simulation output (tier X, pending validation); comparison against Cassini solar-system bounds is in progress and no numerical Cassini parameter is reported here, as none exists yet in our simulation output. Separately, we formalize the Weak Energy Condition $w(t) \geq -1$ as a runtime boundary condition (tier C): when the target-space imaginary modulus τ_{im} is kept positive via projection during integration, $w(t) \geq -1$ holds by construction of that projection, not by an independent proof from metric positivity.

SciPost Acceptance Criteria Fulfillment We address the acceptance criteria of *SciPost Physics* as follows:

- **Methodological Contribution:** It couples a high-performance Rust PDE solver with discrete algebraic consistency checks via Lean 4 in string theory. The solver itself is not formally verified; Lean validates discrete topological/algebraic invariants derived from its output, not the continuous physics.
- **Reproducibility Status:** All simulation scripts and Lean 4 sources are open-source and publicly archived. Locally, 83 theorems were audited against the project’s axioms; 5 fail this audit (`buscher_involution`, plus four theorems in the `HoloAlg` module, all resting on project-specific axioms), and the remaining 78 pass, though many are true by construction of their own definitions rather than by a derivation of the physics they name. Concerning `buscher_involution` specifically: the axiom `inv_inv` declared in `proofs/BuscherRules.lean`, instantiated at `Nat`, proves `False` in the Lean kernel, so this previously claimed result establishes nothing about Buscher T-duality and we withdraw that claim. Independently, the LeanMaster kernel-checked foundation (v2.2.0) verifies 188 theorems with 0 failing, depending only on Lean’s three standard axioms; these are audit-gate counts on discrete algebraic statements, not simulation outputs. Performance/benchmark figures (throughput, timing, and cost comparisons) are pending hardware verification and are not reported in this letter or the revised manuscript.

- **Community Impact:** It introduces practical tools for Double Field Theory, generalized geometry, and Swampland cosmology, accompanied by a companion scientific software package (`leanflow`).

Correction Notice This submission corrects a manuscript previously deposited as Zenodo record 22683565 (DOI 10.5281/zenodo.22683565, concept record 22683564); the corrected manuscript will be issued as a new version of that same concept record. Relative to that original deposit we withdraw: (i) the claimed formal verification of Buscher T-duality involution, for the reason given above; (ii) all performance/benchmark figures, which are pending hardware verification; and (iii) numerical values that could not be traced to any committed simulation output, including the previously reported metric-breakdown minimum, TDA Mapper node/edge/ β_1 counts, the 1.24×10^{-4} screening factor and the 2.48×10^{-6} Cassini parameter, and the previously reported Lean job and test-suite counts, each replaced above or removed pending reproducible sources.

We believe this paper will resonate deeply with the high-energy theoretical physics, mathematical physics, and computational physics readerships of SciPost.

Thank you for your consideration.

Sincerely,

The Authors

SocrateAI Research Initiative